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```
# Install spaCy (run in Colab if needed)
!pip install spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.4.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.5)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2026.2.25)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.3)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (0.24.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (0.23.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2->spacy) (2.1.2)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (4.0
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (2
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-slim<1.0.0,>=0.3.0-
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 94.4 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy
from spacy import displacy
import pandas as pd

# Load model
nlp = spacy.load('en_core_web_sm')
print('Model loaded successfully')
```

```
Model loaded successfully
```

```
sentences = [
    "The boy plays cricket.",
    "She is reading a book.",
    "The teacher explained the topic clearly.",
    "I went home and completed my homework.",
    "He is tired because he worked all day.",
    "What are you doing?",
    "Where is the station located?",
    "The cake was baked by my mother.",
    "The project was completed successfully.",
    "Artificial intelligence is transforming industries.",
    "Students are preparing for exams seriously.",
    "The dog barked loudly at the stranger.",
    "Although it was raining, we went outside.",
    "He bought a car and drove it home.",
    "The scientist discovered a new element."
]

print("Total sentences:", len(sentences))
```

```
Total sentences: 15
```

```
for sent in sentences:
    doc = nlp(sent)
    data = []
    for token in doc:
        data.append([
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])
    df = pd.DataFrame(data, columns=["Token", "Lemma", "POS", "Dependency", "Head"])
    print("\nSentence:", sent)
    display(df)
```

```
Sentence: The boy plays cricket.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	boy
1	boy	boy	NOUN	nsubj	plays
2	plays	play	VERB	ROOT	plays
3	cricket	cricket	NOUN	dobj	plays
4	.	.	PUNCT	punct	plays

```
Sentence: She is reading a book.
```

	Token	Lemma	POS	Dependency	Head
0	She	she	PRON	nsubj	reading
1	is	be	AUX	aux	reading
2	reading	read	VERB	ROOT	reading
3	a	a	DET	det	book
4	book	book	NOUN	dobj	reading
5	.	.	PUNCT	punct	reading

```
Sentence: The teacher explained the topic clearly.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	teacher
1	teacher	teacher	NOUN	nsubj	explained
2	explained	explain	VERB	ROOT	explained
3	the	the	DET	det	topic
4	topic	topic	NOUN	dobj	explained
5	clearly	clearly	ADV	advmod	explained
6	.	.	PUNCT	punct	explained

```
Sentence: I went home and completed my homework.
```

	Token	Lemma	POS	Dependency	Head
0	I	I	PRON	nsubj	went
1	went	go	VERB	ROOT	went
2	home	home	ADV	advmod	went
3	and	and	CCONJ	cc	went
4	completed	complete	VERB	conj	went
5	my	my	PRON	poss	homework
6	homework	homework	NOUN	dobj	completed
7			PUNCT	punct	went

7	.	.	PUNCT	punct	went

```
Sentence: He is tired because he worked all day.
```

	Token	Lemma	POS	Dependency	Head
0	He	he	PRON	nsubj	is
1	is	be	AUX	ROOT	is
2	tired	tired	ADJ	acomp	is
3	because	because	SCONJ	mark	worked
4	he	he	PRON	nsubj	worked
5	worked	work	VERB	advcl	is
6	all	all	DET	det	day
7	day	day	NOUN	npadvmod	worked
8	.	.	PUNCT	punct	is

```
Sentence: What are you doing?
```

	Token	Lemma	POS	Dependency	Head
0	What	what	PRON	dobj	doing
1	are	be	AUX	aux	doing
2	you	you	PRON	nsubj	doing
3	doing	do	VERB	ROOT	doing
4	?	?	PUNCT	punct	doing

```
Sentence: Where is the station located?
```

	Token	Lemma	POS	Dependency	Head
0	Where	where	SCONJ	advmod	is
1	is	be	AUX	auxpass	located
2	the	the	DET	det	station
3	station	station	NOUN	nsubjpass	located
4	located	locate	VERB	ROOT	located
5	?	?	PUNCT	punct	located

```
Sentence: The cake was baked by my mother.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	cake
1	cake	cake	NOUN	nsubjpass	baked
2	was	be	AUX	auxpass	baked
3	baked	bake	VERB	ROOT	baked
4	by	by	ADP	agent	baked

5	my	my	PRON	poss	mother
6	mother	mother	NOUN	pobj	by
7	.	.	PUNCT	punct	baked

```
Sentence: The project was completed successfully.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	project
1	project	project	NOUN	nsubjpass	completed
2	was	be	AUX	auxpass	completed
3	completed	complete	VERB	ROOT	completed
4	successfully	successfully	ADV	advmod	completed
5	.	.	PUNCT	punct	completed

```
Sentence: Artificial intelligence is transforming industries.
```

	Token	Lemma	POS	Dependency	Head
0	Artificial	artificial	ADJ	amod	intelligence
1	intelligence	intelligence	NOUN	nsubj	transforming
2	is	be	AUX	aux	transforming
3	transforming	transform	VERB	ROOT	transforming
4	industries	industry	NOUN	dobj	transforming
5	.	.	PUNCT	punct	transforming

```
Sentence: Students are preparing for exams seriously.
```

	Token	Lemma	POS	Dependency	Head
0	Students	student	NOUN	nsubj	preparing
1	are	be	AUX	aux	preparing
2	preparing	prepare	VERB	ROOT	preparing
3	for	for	ADP	prep	preparing
4	exams	exam	NOUN	pobj	for
5	seriously	seriously	ADV	advmod	preparing
6	.	.	PUNCT	punct	preparing

```
Sentence: The dog barked loudly at the stranger.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	dog
1	dog	dog	NOUN	nsubj	barked
2	barked	bark	VERB	ROOT	barked
3	loudly	loudly	ADV	advmod	barked

4	at	at	ADP	prep	barked
5	the	the	DET	det	stranger
6	stranger	stranger	NOUN	pobj	at
7	.	.	PUNCT	punct	barked

```
Sentence: Although it was raining, we went outside.
```

	Token	Lemma	POS	Dependency	Head
0	Although	although	SCONJ	mark	raining
1	it	it	PRON	nsubj	raining
2	was	be	AUX	aux	raining
3	raining	rain	VERB	advcl	went
4	,	,	PUNCT	punct	went
5	we	we	PRON	nsubj	went
6	went	go	VERB	ROOT	went
7	outside	outside	ADV	advmod	went
8	.	.	PUNCT	punct	went

```
Sentence: He bought a car and drove it home.
```

	Token	Lemma	POS	Dependency	Head
0	He	he	PRON	nsubj	bought
1	bought	buy	VERB	ROOT	bought
2	a	a	DET	det	car
3	car	car	NOUN	dobj	bought
4	and	and	CCONJ	cc	bought
5	drove	drive	VERB	conj	bought
6	it	it	PRON	dobj	drove
7	home	home	ADV	advmod	drove
8	.	.	PUNCT	punct	bought

```
Sentence: The scientist discovered a new element.
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	scientist
1	scientist	scientist	NOUN	nsubj	discovered
2	discovered	discover	VERB	ROOT	discovered
3	a	a	DET	det	element
4	new	new	ADJ	amod	element
5	element	element	NOUN	dobj	discovered

Sentence Structure Analysis

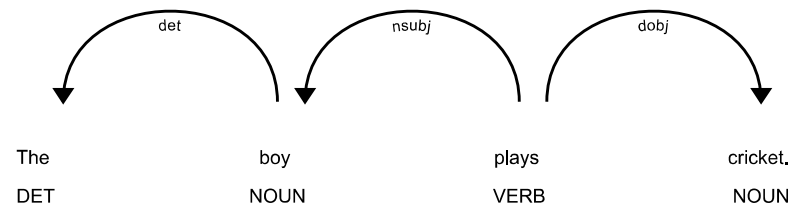
```
def analyze_sentence(sent):
    doc = nlp(sent)
    subject = [token.text for token in doc if "subj" in token.dep_]
    obj = [token.text for token in doc if "obj" in token.dep_]
    root = [token.text for token in doc if token.dep_ == "ROOT"]
    print("Sentence:", sent)
    print("Subject:", subject)
    print("Verb (ROOT):", root)
    print("Object:", obj)
    print("-"*40)

for s in sentences:
    analyze_sentence(s)
```

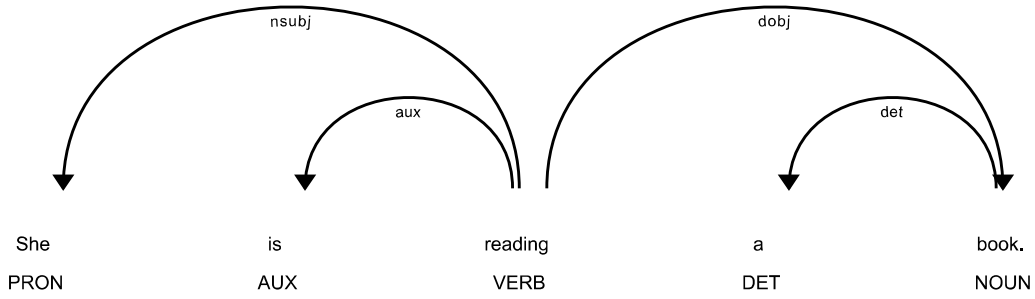
```
Sentence: The boy plays cricket.
Subject: ['boy']
Verb (ROOT): ['plays']
Object: ['cricket']
----------------------------------------
Sentence: She is reading a book.
Subject: ['She']
Verb (ROOT): ['reading']
Object: ['book']
----------------------------------------
Sentence: The teacher explained the topic clearly.
Subject: ['teacher']
Verb (ROOT): ['explained']
Object: ['topic']
----------------------------------------
Sentence: I went home and completed my homework.
Subject: ['I']
Verb (ROOT): ['went']
Object: ['homework']
----------------------------------------
Sentence: He is tired because he worked all day.
Subject: ['He', 'he']
Verb (ROOT): ['is']
Object: []
----------------------------------------
Sentence: What are you doing?
Subject: ['you']
Verb (ROOT): ['doing']
Object: ['What']
----------------------------------------
Sentence: Where is the station located?
Subject: ['station']
Verb (ROOT): ['located']
Object: []
----------------------------------------
Sentence: The cake was baked by my mother.
Subject: ['cake']
Verb (ROOT): ['baked']
Object: ['mother']
----------------------------------------
Sentence: The project was completed successfully.
```

```
Subject: ['project']
Verb (ROOT): ['completed']
Object: []
----------------------------------------
Sentence: Artificial intelligence is transforming industries.
Subject: ['intelligence']
Verb (ROOT): ['transforming']
Object: ['industries']
----------------------------------------
Sentence: Students are preparing for exams seriously.
Subject: ['Students']
Verb (ROOT): ['preparing']
Object: ['exams']
----------------------------------------
Sentence: The dog barked loudly at the stranger.
Subject: ['dog']
```

Dependency Tree Visualization

```
for sent in sentences[:5]:
    doc = nlp(sent)
    displacy.render(doc, style='dep', jupyter=True)
```


det
nsubj
dobj
The
DET
boy
NOUN
plays
VERB
cricket.
NOUN


nsubj
dobj
aux
det
She
PRON
is
AUX
reading
VERB
a
DET
book.
NOUN